

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S) & LIIT **Paper -1(Adv-2019-P1-Model)** **Date: 28-12-2025**

Time: 09.00Am to 12.00Pm **GTA-09** **Max. Marks: 186**

28-12-2025_Sr. XL-500, Sr.S60(N&S) & LIIT_Jee-Adv(2019-P1)_GTA-09_Syllabus,

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

Name of the Student: _____ **H.T. NO:** | | | | | | | | |

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JEE-ADVANCE-2019-P1-Model

Time: 3:00Hr's **IMPORTANT INSTRUCTIONS** Max Marks: 186

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	3	-1	4	12
Sec – II(Q.N : 5 – 12)	Questions with Multiple Correct Choice **+1 partial marks**	4	-1	8	32
Sec – III(Q.N : 13 – 18)	Questions with Numerical Value Answer Type	3	0	6	18
Total				**18**	**62**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 22)	Questions with Single Correct Choice	3	-1	4	12
Sec – II(Q.N : 23 – 30)	Questions with Multiple Correct Choice **+1 partial marks**	4	-1	8	32
Sec – III(Q.N : 31 – 36)	Questions with Numerical Value Answer Type	3	0	6	18
Total				**18**	**62**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 40)	Questions with Single Correct Choice	3	-1	4	12
Sec – II(Q.N : 41 – 48)	Questions with Multiple Correct Choice **+1 partial marks**	4	-1	8	32
Sec – III(Q.N : 49 – 54)	Questions with Numerical Value Answer Type	3	0	6	18
Total				**18**	**62**

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PHYSICS **Max Marks: 62**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct. **Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases. Section 1 (Max Marks: 12)**

- Section 1 contains Four questions
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If the answer is correct)
  Zero Marks: **0** (If the question is unanswered)
  Negative Marks: -1 (In all other cases)

01. For transverse progressive wave: $y=(2cm)\sin(3t-4x)$.

Statement-1: The particle at $x=0$ is moving towards positive y-axis at $t=0$.

Statement-2: The particle at $x=8m$ is in phase with particle at $x=1m$.

Statement-3: The acceleration of particle at $x=\frac{\pi}{8}m$ and at $t=\frac{\pi}{6}\sec$ is increasing in magnitude.

Statement-4: The velocity of particle at $x=\frac{\pi}{6}m$ is in direction of propagation of wave at all times.

The CORRECT statement(s) are:

A) 1 **B)** 1 and 3 **C)** 1,2 and 3 **D)** 1,2,3 and 4

02. For a liquid moving in a cylindrical tube of radius R and length l, the pressure difference at the ends of the tube is related by poiseuille's law given by $\Delta P=\left(\frac{8\eta l}{\pi r^4}\right)Q$. For a tube having radius $r=2.00cm\pm0.01cm$, $l=10.0cm\pm0.1cm$ the error in measurement of pressure difference is: (Take: $\eta=\pi\,N-s/m^2, Q=1m^3/\sec$ exactly)

A) 2% **B)** 3% **C)** 4% **D)** 6%

03. A thin mica sheet $(\mu=1.6)$ is used to cover one slit of Young's double slit arrangement. The central fringe is now formed at the previous seventh bright fringe. If $\lambda=5500\,\text{A}^0$, the thickness of the mica sheet is:

A) $1.2\mu m$ **B)** $6.41\mu m$ **C)** $6.4mm$ **D)** $2.4\mu m$

04. Two identical photons of energy $E(4eV\le E\le 8eV)$ fall on two different metals whose work functions are in the ratio of 1: 2. The ratio of the kinetic energies of the most energetic electrons coming from each metal is 2:1. Work function of one of the metal is 4 eV. What is the energy of photon in eV.

A) 6 eV **B)** 8 eV **C)** 4 eV **D)** 5 eV

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SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct. **Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -1 in all wrong cases**

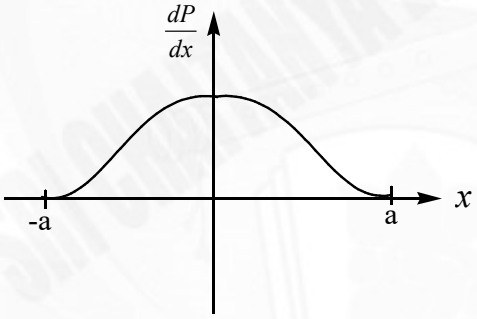
Section 2 (Max Marks: 32)

- Section two contains Eight (8) questions
- Each Question has Four Options and One Or More Than One of these four will be the correct answer.
- For each question, choose the option(s) corresponding to all the correct answer(s)

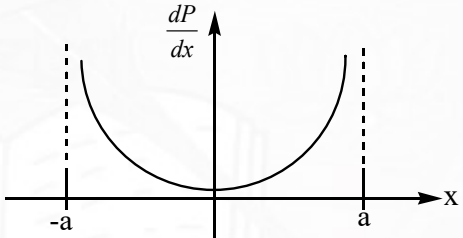
The Marking scheme to evaluate Answer to each question will be :
Full Marks: **+4** (If the answer is correct)
Partial Marks: **+3** (If all the four options are correct but **only 3** are chosen)
Partial Marks: **+2** (If THREE or More options are correct but **only 2** correct answers are chosen)
Partial Marks: **+1** (If TWO or More options are correct but **only 1** correct answer is chosen)
Zero Marks: **0** (If the question is unanswered)
Negative Marks: -1 (In all other cases)

05. A body is executing SHM on a straight line (x-axis) with time period 2sec and amplitude 1cm. assume that probability of the body to be from $x=-1cm$ to $x=+1cm$ is 1. Probability of finding particle within interval x to $x+dx$ is dP. Probability of particle having velocity from v to $v+dv$ is dP. Choose the correct plot(S). of $\frac{dP}{dx}$ and $\frac{dP}{dv}$ against x and v respectively

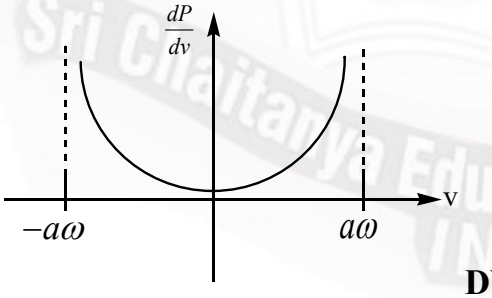
A)



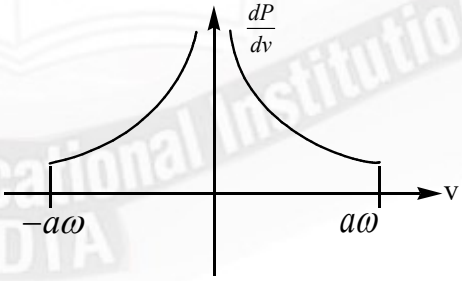
B)



C)



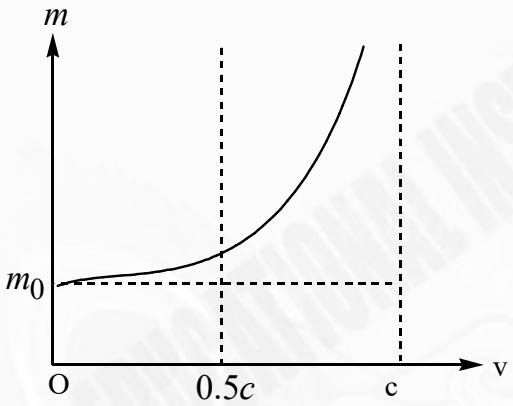
D)



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06. Mass $m = \dfrac{m_0}{\sqrt{1-\dfrac{v^2}{c^2}}}$ as a function of velocity is shown in the graph. m_0 is the rest mass of the system. The kinetic energy of system of given as $K = mc^2 - m_0c^2$.

Choose correct option(s)



A) If a system is moving with speed $v = 0.5c$, the mass of system is increased by 15.5%.

B) Electrons in Cornell university synchrotron reach a velocity of $v = 0.8c$, The mass of these electrons are $\left(\dfrac{5}{3}\right)$ time the rest mass of electron.

C) The Bevatron a proton accelerator generates the accelerated protons with kinetic energy 10^{-9} *Joules*. The mass of accelerated protons is 8.68 times the actual mass of proton approximately $\left(m_{p_0} = 1.67 \times 10^{-27}\, kg\right)$.

D) If the kinetic energy of particle is equal to its rest mass energy then speed of particle is $0.75c$ approximately.

07. In a meter bridge experiment with the resistance R_1 in the left gap and a resistance X in the right gap, the null point obtained at 40cm from left end. Now with the resistance R_2 in the left gap and the same resistance X in the right gap, null point is obtained at 50cm from left end. (Ignore end correction)

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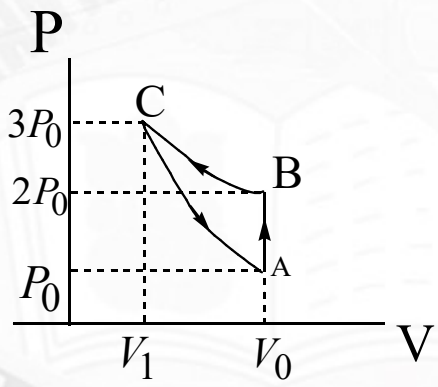
A) If R_1 *and* R_2 are put in series in the left gap and the right gap contains resistance X, the null point is obtained at a distance 75 cm from the left end.

B) If R_1 *and* R_2 are put in series in the left gap and the right gap contains resistance X, the null point is obtained at a distance $\frac{125}{2}cm$ from the left end.

C) If R_1 *and* R_2 are put in parallel in the right gap and the left gap contains resistance X, the null point is obtained at a distance $\frac{500}{7}cm$ from the left end.

D) If R_1 *and* R_2 are put in parallel in the left gap and the right gap contains resistance X, the null point is obtained at a distance $\frac{200}{7}cm$ from the left end.

08. One mole of an ideal gas undergoes a thermodynamics cyclic process as shown in the figure. The cyclic process consists of an isochoric, an adiabatic process and an isothermal (may not be in the same sequence).



A) The ratio of molar specific heat at constant pressure to that at constant volume is $\frac{ln3}{ln3-ln2}$.

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B) The ratio of molar specific heat at constant pressure to that at constant volume is $\frac{ln2}{ln3-ln2}$.

C) Magnitude of work done on the gas in the isothermal process is $2P_0V_0\ ln(3/2)$.

D) Magnitude of work done by the gas in the isothermal process is $3P_0V_0\ ln(3/2)$.

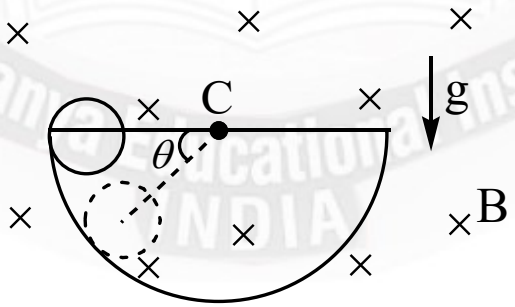
09. Conducting shell of radius a is surrounded by another spherical conducting shell of radius b. The space between them is filled with dielectric material of conductivity σ and dielectric constant K. The charge Q_1 *and* Q_2 are given to the inner and outer shell at time $t=0$ respectively. Charge on outer shell at time t is

A) $Q_1+Q_2\left(1-\exp\left(-\frac{\sigma t}{\varepsilon_0 K}\right)\right)$ **B)** $Q_2-Q_1\left(1-\exp\left(-\frac{\sigma t}{\varepsilon_0 K}\right)\right)$

C) $Q_1-Q_2\left(1-\exp\left(-\frac{\sigma t}{\varepsilon_0 K}\right)\right)$ **D)** $Q_2+Q_1\left(1-\exp\left(-\frac{\sigma t}{\varepsilon_0 K}\right)\right)$

10. A solid sphere is held at rest on the fixed hemispherical bowl made of insulating material such that centre of sphere lies on the horizontal diameter as shown in figure. Total charge $q=5.1\times10^{-1}C$ and mass m = 42 grams, both are distributed uniformly in the sphere. There exists uniform magnetic field B in space as shown in figure. Consider contact surface to be sufficient rough. Spherical bowl is made of non-magnetic material. Radius of bowl = 10 cm, radius of sphere 3 cm. $g=10\,m/s^2$.

Now if sphere is released



Fixed bowl

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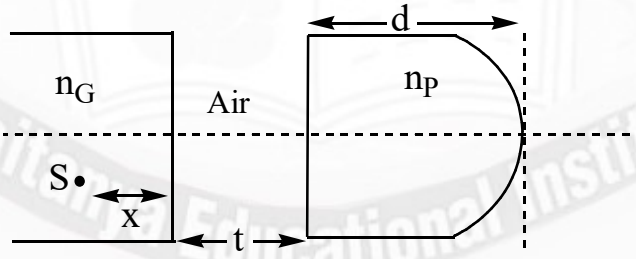
A) At an angle θ with horizontal, velocity of COM of sphere will be $v=\sqrt{\frac{10}{7}g(R-r)\sin\theta}$.

B) At an angle θ with horizontal, velocity of COM of sphere will be $v=\sqrt{\frac{8}{7}g(R-r)\sin\theta}$.

C) The limiting value of B (approximately) so that on releasing sphere, it does not leave contact with bowl in course of motion will be 2T.

D) The limiting value of B (approximately) so that on releasing sphere, it does not leave contact with bowl in course of motion will be 4T.

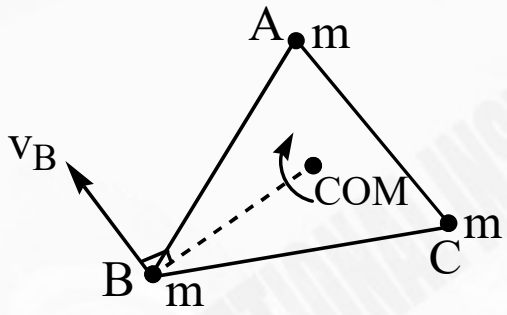
11. As shown below, a point light source (S) is inside a glass block of refractive index $n_G=\frac{3}{2}$, at a distance 2 mm from the optical axis and distance x = 3 cm from the flat surface. A thick plano-convex lens made of transparent plastic material of refractive index $n_P=2$, is placed near the source as shown. The thickness of the lens is d, and the radius of convex surface is R = 15 cm. The width of air gap between the glass block and the lens is 't'. The width of air gap t and thickness of lens d is adjusted so that light (paraxial rays) from point source forms always a parallel beam after passing through the lens. Select correct option(s).



A) For d = 16cm, width of air gap t = 5 cm.
B) For thickness d = 10cm, width of air gap t = 8cm.
C) The maximum adjustable value of d = 26 cm.
D) The maximum adjustable value of d = 16cm.

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12. Three small balls A, B and C of same mass , are fixed by massless rods at the vertices of equilateral triangle with side L. The system is placed on a smooth horizontal surface and set in rotation about the stationary COM with angular speed ω. At a certain instant ball B breaks off the system with a velocity $\vec{v}_B$ as shown in diagram where $|\vec{v}_B| = \frac{2\omega L}{\sqrt{3}}$.



A) The COM of A and C moves with a velocity $\frac{-\vec{v}_B}{2}$.

B) The tension in rod connecting A and C after B breaks off is $m\omega^2\frac{\ell}{2}$.

C) The speed of A just after B breaks off is $\frac{v_B}{2}$.

D) The speed of A, after a time $\frac{\pi}{\omega}$ after B breaks off is $\frac{\sqrt{19}}{6}\omega\ell$.

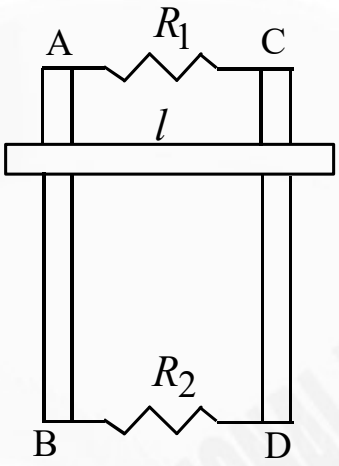
SECTION – III
(NUMERICAL VALUE TYPE)

This section contains 6 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. **(e.g. 003.2343, 6.250023, 7.000000, -0.33, -.30, 30.27000, -127.0030).**
Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases. Section 3 (Max Marks: 18)

- This Section contains **Six (06)** Questions. The answer to each question is a Numerical Value
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If ONLY the correct numerical value is entered)
  Zero Marks: 0 (In all other cases)

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13. Two parallel vertical metallic rails AB and CD are separated by 1m. They are connected at two ends by resistances R_1 and R_2 as shown in figure.



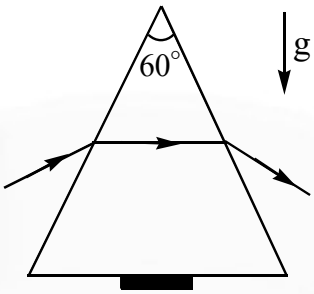
A horizontal metallic bar L of mass 0.2kg slides without friction vertically down the rails under the action of gravity starting from rest. There is a uniform horizontal magnetic field of 0.6 tesla perpendicular to the plane of the rails. It is observed that when the terminal velocity is attained, the powers dissipated in R_1 and R_2 are 0.76 watt and 1.2 watt respectively. Find the terminal velocity (in m/s) of the bar L. (neglect resistance of the bar and take $g = 9.8 m/s^2$)

14. A glass rod of diameter $d_1 = 1.5mm$ is inserted symmetrically into a glass capillary with inside diameter $d_2 = 2.0mm$. Then the whole arrangement is vertically oriented and brought in contact with the surface of water. To what height will the water rise in the capillary in cm($\sigma_w = 0.075 N/m, \rho_w = 10^3 Kg/m^3, g = 10 m/s^2$)

15. An equilateral glass prism kept on a table has refractive index of $\mu = \sqrt{2}$. It is illuminated by a narrow laser beam having power P_0 and wavelength λ. The path of the laser beam inside the prism is parallel to the base of the prism. The change in reading of weight of the prism due to the incident laser beam is $\frac{P_0}{c}\left(\frac{\sqrt{x}-1}{\sqrt{y}}\right)$.

Find $x + y$? c is the speed of light. (Ignore any partial reflection)

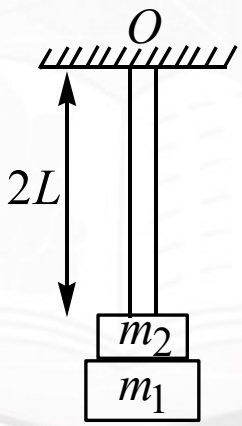
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Weighing machine

16. A body is launched from the earth's surface at an angle $\alpha = 30^\circ$ to the horizontal at a speed $v_0 = \sqrt{\frac{1.5GM}{R}}$. Neglecting air resistance and earth's rotation. The height from earth surface to which the body will rise is found to be K times 0.29R. Find K.
(here M is mass of earth and R the radius of earth)

17. A string of negligible mass, natural length L has Young's modulus Y. The string hangs from roof with mass m_1 and m_2 as shown in figure. If mass m_2 is removed gently, the mass m_1 is just able to bounce back upto point O. Find the ratio m_2 / m_1.
(string will not obstruct the motion of mass m_1 and system is initially in equilibrium).
[m_2 is placed on m_1 which is tied to the string and Hooke's is obeyed]



18. Use the following equation $(m)^{ft}(a-c)^2 = x\left[d - kx - \frac{mc}{t}\right]$; where k = spring constant, x = distance, m = mass, t = sec and a, c, d and f are some other physical quantities. If the dimensional formula of f is $\left[M^{\ell}L^{m}T^{-n}\right]$, then the value of $\ell + m + n$ is

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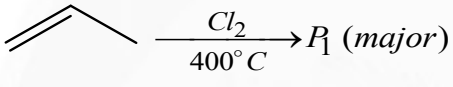
CHEMISTRY **Max Marks: 62**

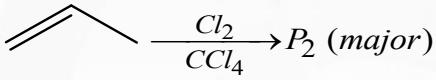
SECTION – I
(SINGLE CORRECT ANSWER TYPE)

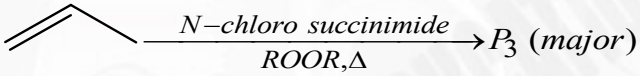
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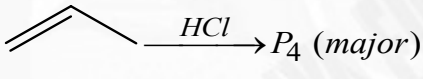
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  Zero Marks: **0** (If the question is unanswered)
  Negative Marks: -1 (In all other cases)

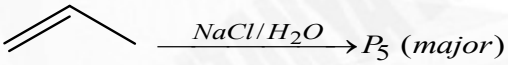
19. Select the incorrect statements about products.

i) 

ii) 

iii) 

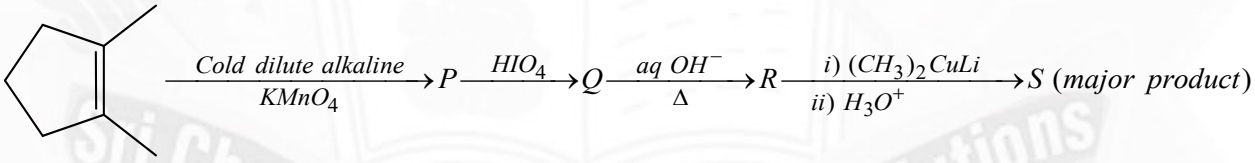
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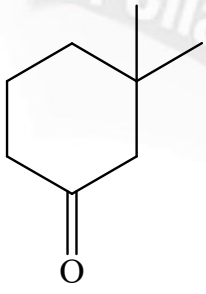
v) 

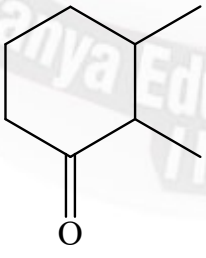
A) P_1 and P_3 are identical **B)** P_4 and P_5 are identical

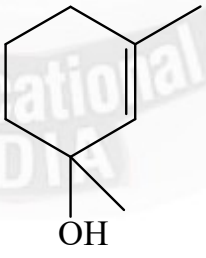
C) P_2 is dihalide **D)** P_3 is a mono halide

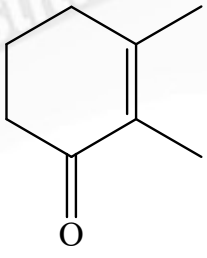
20. In the following reaction sequence, product S is



A) 

B) 

C) 

D) 

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21. The reaction: $2NO(g)+2H_2(g)\rightarrow N_2(g)+2H_2O(g)$ has been assigned to follow the following mechanism

I. $NO+NO \rightleftharpoons N_2O_2(fast)$

II. $N_2O_2+H_2\rightarrow N_2O+H_2O\ (slow)$

III. $N_2O+H_2\longrightarrow N_2+H_2O\ (fast)$

The rate constant of step II is $1.2\times10^{-4}\,mole^{-1}L.\min^{-1}$. While equilibrium constant of step I is 1.4×10^{-2}. What is the rate of reaction when concentration of NO and H_2 each is $0.5\ mole\ L^{-1}$.

A) $2.1\times10^{-7}\,molL^{-1}\min^{-1}$ **B)** $3.2\times10^{-6}\,mol\,L^{-1}\min^{-1}$

C) $3.5\times10^{-4}\,mol\,L^{-1}\min^{-1}$ **D)** $3.2\times10^{-5}\,mol.L^{-1}\min^{-1}$

22. Which one of the following sets contains only non-polar and planar molecules/ions and also use d-orbitals in their bonding?

A) $XeF_2, ClF_3, XeF_4, SO_4^{2-}$ **B)** $ClF_3, XeO_3, XeO_4, SO_2$

C) $XeO_4, BrF_5, HClO_4, XeO_3F_2$ **D)** $XeF_2, XeF_4, ICl_4^-, SO_3$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct. **Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -1 in all wrong cases**

Section 2 (Max Marks: 32)

- Section two contains Eight (8) questions
- Each Question has Four Options and One Or More Than One of these four will be the correct answer.
- For each question, choose the option(s) corresponding to all the correct answer(s)

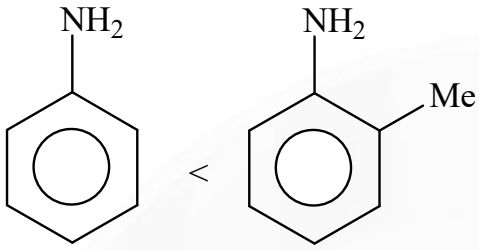
The Marking scheme to evaluate Answer to each question will be :
Full Marks: **+4** (If the answer is correct)
Partial Marks: **+3** (If all the four options are correct but **only 3** are chosen)
Partial Marks: **+2** (If THREE or More options are correct but **only 2** correct answers are chosen)
Partial Marks: **+1** (If TWO or More options are correct but **only 1** correct answer is chosen)
Zero Marks: **0** (If the question is unanswered)
Negative Marks: -1 (In all other cases)

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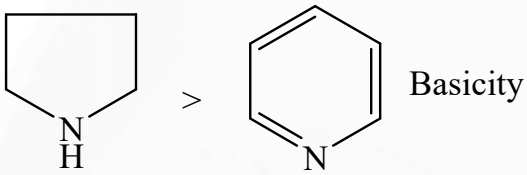

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23. Select correct options:

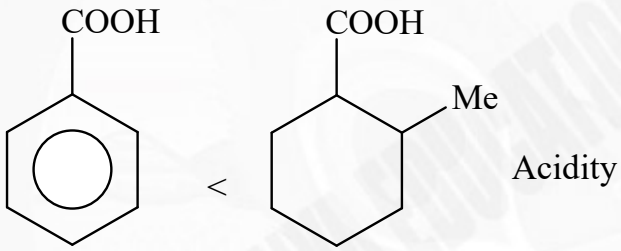
A)



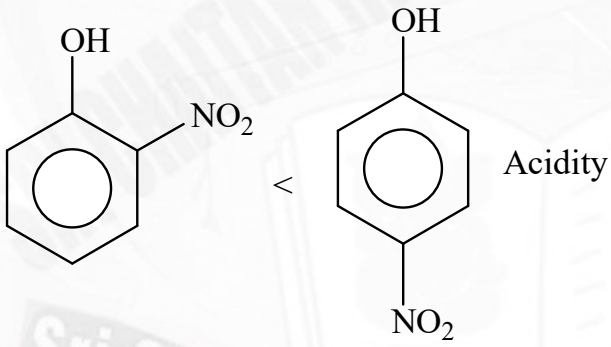
B)



C)

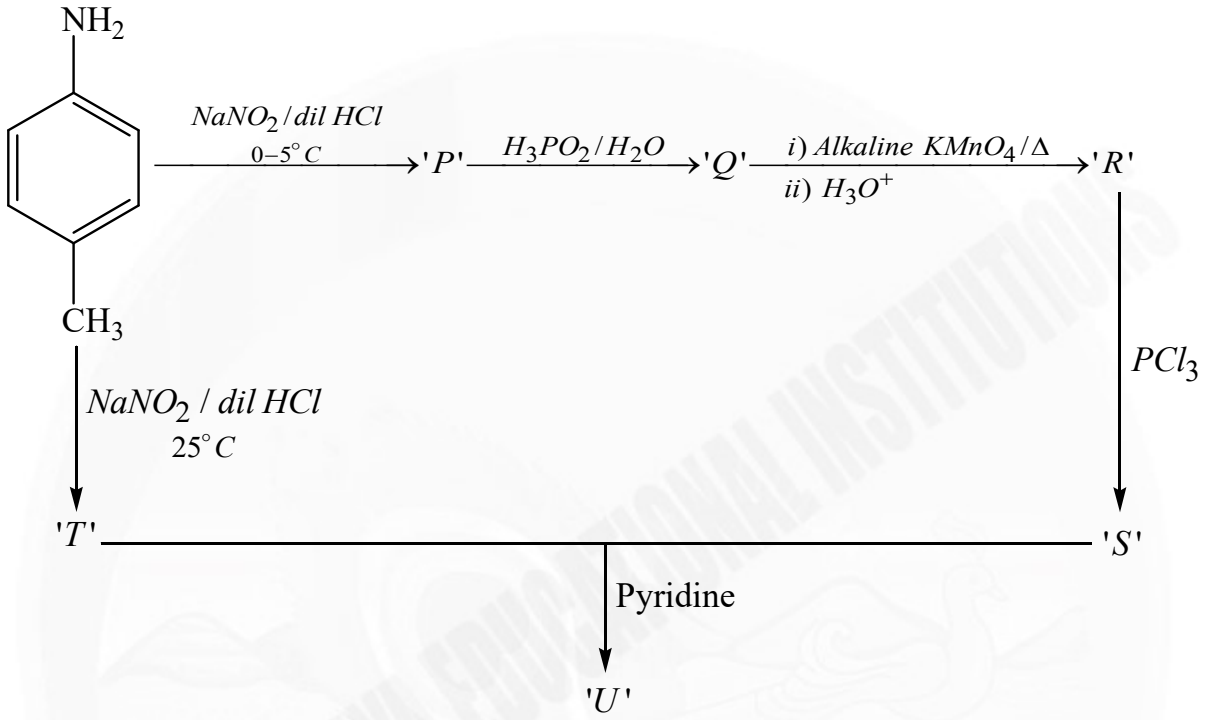


D)



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24. how many of the following is/are incorrect regarding below formed compounds?



A) Compound T will give phthalein test.

B) Compound T will give neutral $FeCl_3$ test.

C) Compound R upon fridalcrafts acylation gives meta as the major product.

D) Compound U contains three tertiary carbons.

25. Which of the following statement(s) is/are correct regarding the commercial extraction of Al and Fe from their oxide ores?

A) Concentration of bauxite ore is carried out by heating the powdered ore with a concentrated solution of NaOH at 473 – 523K and 35 – 36 bar pressure.

B) Freshly prepared hydrated Al_2O_3 act as seeding reagent in one of the step during the extraction of Al.

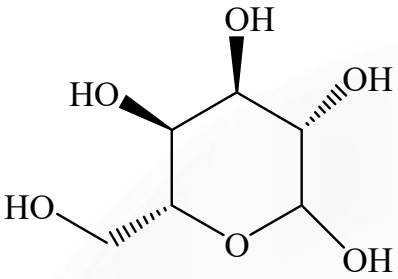
C) At the lower temperature range (500 – 800K) in the blast furnace (used for the extraction of iron), CO function as a reducing agent.

D) Major slag removed during the extraction of pig iron is $CaSiO_3$.

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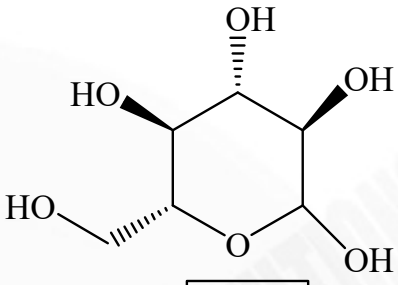
26. Which of the following is/are showing correct structures for the given names?

A)



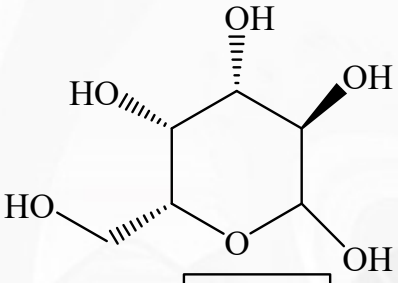
Allose

B)



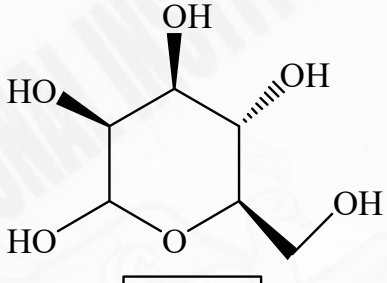
Glucose

C)



Galactose

D)



Mannose

27. If Cl_2 and PCl_3 can react to form PCl_5 according to following reaction

$Cl_2 + PCl_3 \rightarrow PCl_5$

In tube A, 355gm Cl_2 is taken and in tube B, 1375gm PCl_3 is taken. When half the mass of gas in tube A is transferred to B, reaction took place in B. After the reaction is complete, half of the mass of mixture in B is again transferred to A to cause same reaction in container A. (no reversibility)

Select the correct option(s).

A) Tube A contains 3.75 moles of PCl_5 & 1.25 moles of PCl_3 finally

B) Tube A contains 2.5 moles of Cl_2 & 1.25 moles of PCl_5 finally

C) Tube B contains 3.75 moles of PCl_3 & 1.25 moles of PCl_5 finally

D) Tube B contains equal number of moles PCl_3 & PCl_5 finally

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28. Assume ideal gas behaviour for all the gases & vibrational degrees of freedom to be active. Separate equimolar samples of Ne, O_2, SO_2 & CH_4 were subjected to a two step process as mentioned. Initially all are at same state of temperature & pressure.

Step I $\rightarrow$ All undergo reversible adiabatic expansion to attain same final volume, which is double the original volume thereby causing the decreases in their temperature.

Step I $\rightarrow$ After step I all are given appropriate amount of heat isochorically to restore the original temperature.

Mark the correct option(s).

A) Due to step I only, the decrease in temperature will be maximum for Ne.

B) During step II, heat given will be minimum for CH_4.

C) There will be no net change in internal energy for any of the gas after both the steps of process are completed.

D) The P-V graph of CH_4 & SO_2 will be same for overall process.

29. Which of the following can discharge the pink colour of $KMnO_4$ in acidic medium

A) Mohr's salt **B)** Oxalic acid **C)** Ferric sulphate **D)** H_2O_2

30. Which of the following statement(s) is/are incorrect?

A) $E^0_{Mn^{3+}/Mn^{+2}}$ value is much more positive than $E^0_{Co^{3+}/Co^{2+}}$ $\left(E^0_{M^{3+}/M^{+2}} : SRP\ value\right)$

B) In 3d series, $E^0_{M^{2+}/M}$ value is highest for Cu^{2+}/Cu couple

C) Commercially permanganate is prepared by the alkaline oxidative fusion of MnO_2 followed by the electrolytic reduction of manganate

D) Both manganate and permanganate ions are tetrahedral

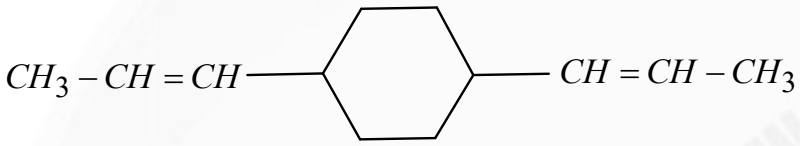
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SECTION – III
(NUMERICAL VALUE TYPE)

This section contains 6 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. **(e.g. 003.2343, 6.250023, 7.000000, -0.33, -.30, 30.27000, -127.0030).**
Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases. Section 3 (Max Marks: 18)

- This Section contains **Six (06)** Questions. The answer to each question is a Numerical Value
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If ONLY the correct numerical value is entered)
  Zero Marks: 0 (In all other cases)

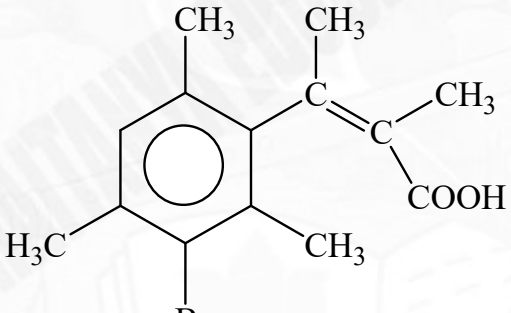
31.



(I)



(II)



(III)

If X is the total number of stereo isomers possible for the compound I **&** Y is the total number of stereo isomers possible for the compound II & Z is the total number of stereo isomers possible for the compound III.
Find the value of X+Y+Z?

32. Acid sample is prepared using HCl, H_2SO_4, H_2SO_3 & H_3PO_4 separately or as mixture of any two or more. Calculate the minimum volume of 4% w/v $NaOH$ added (in litre) to 294gm sample, in order to ensure complete neutralisation in every possible case.

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33. Calculate $[P^{OH}]$ for the following reaction taking place under basic medium at 300K, when the equilibrium is reached

$I_2(s) \rightleftharpoons I^-(aq) + IO_3^-(aq)$

Given:

Substance	$I^-(aq)$	$IO_3^-(aq)$	$H_2O(1)$	$OH^-(aq)$
$\Delta G_f^o(in\ kJ\ mol^{-1})$	–50	–123.5	–233	–150

Equilibrium concentration of $I^- = 0.1M$

Equilibrium concentration of $IO_3^- = 0.1M$

$R = \frac{25}{3} J\,mol^{-1}K^{-1}; \ln 10 = 2.3$

34. A sparingly soluble salt MX is dissolved in water to prepare 1L saturated solution. Now 10^{-6} mole NaX (assume 100% dissociation) is added into this. Conductivity of this solution is $29\times10^{-6}\ S/m$. If K_{sp} of MX is $a\times10^{-b}$, then find value of $(a+b)$?

Given : $\lambda^0_{X^-} = 4\times10^{-3}\ S\ m^2\ mol^-$

$\lambda^0_{Na^+} = 5\times10^{-3}\ S\ m^2\ mol^-$

$\lambda^0_{M^+} = 6\times10^{-3}\ S\ m^2\ mol^-$

35. Nitric oxide emitted from the exhaust systems of supersonic jet aeroplanes is slowly depleting the concentration of the ozone layer in the upper atmosphere. The amount (in grams) of diatomic gas obtained when 60 grams of nitric oxide react with 96g of ozone is ____ .

36. **(i)** $CuSO_4\ solution \xrightarrow[excess]{NH_3\ solution} complex - I$

(ii) $ZnSO_4\ solution \xrightarrow[excess]{NaOH\ solution} complex - II$

Then sum of total number of unpaired electrons present in complex – I and complex – II is ____ .

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MATHEMATICS **Max Marks: 62**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct. **Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases. Section 1 (Max Marks: 12)**

- Section 1 contains Four questions
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If the answer is correct)
  Zero Marks: **0** (If the question is unanswered)
  Negative Marks: -1 (In all other cases)

37. Consider $P: y^2 = 4ax$ and $RH: x^2 - y^2 = a^2$. Let $S = \{m \in R | m$ is the slope of the normal to P that touches the RH$\}$. Then $|S|$ is___ (where $|M|$ denotes the number of element(s) in the set M)

A) 0 **B)** 2 **C)** 4 **D)** 6

38. If $f: R \to R \quad f(x) = |AB|$, where $A_{3\times 2} = \begin{bmatrix} x+1 & x+2 \\ 3x+3 & x+5 \\ 6x+2 & x+4 \end{bmatrix}; B_{2\times 3} = \begin{bmatrix} x+1 & x+2 & 5x+7 \\ x+2 & x+7 & 8x \end{bmatrix}$.

Then the value of $f(1) - 2f'(1) + f''(1) =$ _____
(where $|M|$ denotes determinant of square matrix M)

A) 0 **B)** 3 **C)** $\frac{5}{2}$ **D)** 4

39. Let $g(x)$ be a polynomial of degree one and consider a continuous function $f(x)$ satisfying $f'(1) = f(-1)$ defined as $f(x) = \begin{cases} g(x), & x \le 0 \\ \left(\frac{1+x}{2+x}\right)^{1/x}, & x > 0 \end{cases}$, then which of the following is true

A) $f(0) = 1$ **B)** $f(-1) = \frac{1}{6} + \ln\left(\frac{3}{2}\right)$

C) $f\left(-\frac{3}{2}\right) = \frac{1}{6} + \ln\left(\frac{3}{2}\right)$ **D)** $f\left(-\frac{2}{3}\right) = \left(\frac{1}{6} + \ln\left(\frac{3}{2}\right)\right)$

40. If $\sin^{-1}\left(\sin\frac{33\pi}{7}\right) + \cos^{-1}\left(\cos\left(\frac{46\pi}{7}\right)\right) + \tan^{-1}\left(-\tan\frac{13\pi}{8}\right) + \cot^{-1}\left(\cot\left(-\frac{19\pi}{8}\right)\right) = \frac{a\pi}{b}$

where a and b are co prime positive integers, then the value of $(a+b)$ is

A) 17 **B)** 20 **C)** 23 **D)** 25

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SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct. **Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -1 in all wrong cases**

Section 2 (Max Marks: 32)

- Section two contains Eight (8) questions
- Each Question has Four Options and One Or More Than One of these four will be the correct answer.
- For each question, choose the option(s) corresponding to all the correct answer(s)

The Marking scheme to evaluate Answer to each question will be :
Full Marks: **+4** (If the answer is correct)
Partial Marks: **+3** (If all the four options are correct but **only 3** are chosen)
Partial Marks: **+2** (If THREE or More options are correct but **only 2** correct answers are chosen)
Partial Marks: **+1** (If TWO or More options are correct but **only 1** correct answer is chosen)
Zero Marks: **0** (If the question is unanswered)
Negative Marks: -1 (In all other cases)

41. Let $f:(0,\infty)\to R$ be a differentiable function satisfying the equation

$$f(xy)=e^{xy-x-y}\left(e^{y}f(x)+e^{x}f(y)\right)\ \forall\, x,y>0\,.$$

If $f'(1)=e$, then which of the following is/are **CORRECT**?

A) $\lim\limits_{x\to e}\left[\dfrac{f(x)-e^{x}}{x-e}\right]=e^{(e-1)}$

B) Number of distinct solution(s) of the equation $f(x)=xe^{x}$ in $(0,\infty)$ is 2

C) $\int\limits_{1}^{e} f(x)dx<e^{e}(e-1)$

D) $f(x)$ is a strictly increasing function in $(0,\infty)$

42. Let S be the set of all 3×3 matrices having 3 entries equal to 1 and 6 entries equal to 0. A matrix M is picked uniformly at random from the set S. Then, which of the following is/are **CORRECT**?

A) Total number of matrices in the set is 84

B) Probability that M is non-singular $=\dfrac{1}{14}$

C) Probability that M is identity matrix $=\dfrac{1}{14}$

D) Probability that M has trace equal to zero $=\dfrac{5}{21}$

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43. If $f(P)$ is the number of common tangent lines of two parabolas $x^2 = 2y$ & $\left(y+\frac{1}{2}\right)^2 = 4Px$, then which of the following is/are **CORRECT**?

A) $f(P)=1$ if $P \in \left(-\infty, \frac{-1}{3\sqrt{3}}\right)$ **B)** $f(P)=2$ if $P \in \left(-\frac{1}{3\sqrt{3}}, \frac{1}{3\sqrt{3}}\right)$

C) $f(P)=3$ if $P \in \left(-\frac{1}{3\sqrt{3}}, \frac{1}{3\sqrt{3}}\right)$ **D)** $f(P)=4$ if $P \in \left(\frac{1}{3\sqrt{3}}, \infty\right)$

44. $P(x)=1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\ldots+\frac{x^n}{n!}$, then which of the following option(s) is/are **INCORRECT**?

A) For $n=2021$, $y=P(x)$ has exactly 1 real root

B) For $n=2022$, $y=P(x)$ has exactly 2 real roots

C) For $n=3; y=P(x)+P'(x)$ is always strictly increasing

D) For $n=4; y=P(x)+P'(x)$ is always strictly increasing

45. If $(4+\sqrt{15})^n = I+f$, where n is an odd natural number, I is an integer and $1<f<2$, then which of the following option(s) is/are correct?

A) I is odd integer **B)** I is an even integer

C) $(I+f)(1-f)=1$ **D)** $1-f=(4-\sqrt{15})^n$

46. Let P_0 be an equilateral triangle of area 10sq.units. Each side of P_0 is trisected into three segments of equal length, and the corners of P_0 are snipped off, creating a new polygon (in fact, a hexagon) P_1. Now repeat the process to P_1 – i.e. trisect each side and snip off the corners – to obtain a new polygon P_2. Now repeat this process infinitely to create an object P_∞. Then the area of $P_\infty = \frac{m}{n}$ sq.units, where $m,n \in N$ and m, n are coprime to each other. Then which of the following options is/are incorrect?

A) $m-3n \geq 25$ **B)** $3m-n \leq 131$ **C)** $m-n \leq 30$ **D)** $m+n \leq 32$

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47. Two Arithmetic progressions named as A_1 and A_2 having same number of terms equal to k. The ratio of the last term of the A_1 to the first term of A_2 equals the ratio of the last term of the A_2 to the first term of the A_1, both of which are equal to 4. The ratio of the sum of all k terms of A_1 to the sum of all k terms of A_2 is equal to 2. Let α be the ratio of common difference of the series A_1 to series A_2. Let λ be the ratio of the kth term of the A_1 to A_2 .

(consider the ratio of a to b meaning as $\frac{a}{b}$).Then which of the following option(s) is/are **CORRECT**?

A) The value of α is 26

B) The value of α is 28

C) The value of λ is $\frac{7}{2}$

D) The value of $\alpha+2\lambda$ is 35

48. Two teachers A and B write questions and corresponding solutions for a mathematics test. A writes 10 questions every hour but makes the mistakes in 10% of the solutions. B writes 20 questions per hour and makes the mistakes in the 20% of the solutions, Both of them work for 3 hours and send all the questions to teacher C to be checked. However, teacher C is not really so smart and only 75% of the questions he thinks are wrong are actually incorrect.

Teacher C thinks 20% of questions from A have incorrect solutions and that 10% of questions from B have incorrect solutions. Let the following events be defined as

W: solution to a problem is incorrect

S: teacher C thinks solution is incorrect

E_1: Problem is formed by A

E_2: Problem is formed by B

Then which of the following option(s) is/are **CORRECT**?

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JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
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A) The probability that solution to a randomly selected problem is correct, i.e, $P(\bar{W})$ is equal to $\frac{5}{6}$

B) The probability that teacher C thinks a given problem has correct solution, i.e $P(\bar{S})$ is equal to $\frac{13}{15}$

C) If solution of a problem randomly selected is definitely correct, then the probability that it was formed by A, i.e, $P\left(\frac{E_1}{\bar{W}}\right)$ is equal to $\frac{13}{25}$

D) If solution of a problem randomly selected is definitely correct, then the probability that it was formed by A, i.e, $P\left(\frac{E_1}{\bar{W}}\right)$ is equal to $\frac{9}{25}$

SECTION – III
(NUMERICAL VALUE TYPE)

This section contains 6 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. **(e.g. 003.2343, 6.250023, 7.000000, -0.33, -.30, 30.27000, -127.0030).**
Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases. Section 3 (Max Marks: 18)

- This Section contains **Six (06)** Questions. The answer to each question is a Numerical Value
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If ONLY the correct numerical value is entered)
  Zero Marks: 0 (In all other cases)

49. Let N be the number of words which can be formed using all the letters of the word 'DARJEELING' so that there are at least two consonants between any two vowels. Then the sum of digit(s) of the sum of digit(s) of N is ___

50. For any sequence of real numbers which is defined as $A=(a_1,a_2,a_3,.....)$, a sequence ΔA is defined such that $\Delta A=(a_2-a_1,a_3-a_2,a_4-a_3,.....)$. Suppose that all of the terms of the sequence $\Delta(\Delta A)$ are 1 and that $a_{19}=a_{92}=0$. Then the value of a_3 is ___

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51. Let x be a real number such that $\sin^{10} x + \cos^{10} x = \frac{11}{36}$. Then $\sin^{12} x + \cos^{12} x = \frac{m}{n}$, where m and n are relatively prime positive integers. Then the value of $m+n$ is __

52. In a college of SRI CHAITANYA AKSHARA BHAVAN, all logarithms whose bases are not specified are assumed to be base b, for some fixed $b \geq 2$. A student from that college writes down the equations

$3\log\left(\sqrt{x}\log x\right) = 56$,

$\log_{\log x}(x) = 54$

and finds that this system of equations has a single real number solution $x > 1$. Then the value of b is ___

53. Let a function $f : \mathbb{R} \to \mathbb{R}$ such that:

$f(x)f(y) = f(x+y) + \left(\frac{2025}{2026}\right)^2 .xy,\ \forall x, y \in \mathbb{R}$. For all such possible functions, the value of the sum of all distinct possible values of $f(2026)$ is ___

54. The standard deviation of the following distribution is S,

Class Interval	0-500	500-1000	1000-1500	1500-2000	2000-3000
Frequency	90	218	86	41	15

Then the value of $[S]$ is ___, where $[.]$ is G.I.F.

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APP. NO. 250310255592
CLASSROOM STUDENT FROM GRADE IX - XII



DEVDUTTA MAJHI
APP. NO. 250310016185
DLP/AITS STUDENT



TOSHNIWAL SHIVEN
APP. NO. 250310391420
DLP/AITS STUDENT



SAKSHAM JINDAL
APP. NO. 250310236696
DLP/AITS STUDENT

BELOW 100 ALL INDIA OPEN CATEGORY RANKS	BELOW 500 ALL INDIA OPEN CATEGORY RANKS	BELOW 10 ALL INDIA CATEGORY RANKS COUNT	BELOW 100 ALL INDIA CATEGORY RANKS COUNT	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT
31	95	10	98	579

TOTAL QUALIFIED RANKS FOR JEE ADVANCED-2025 22,094

*DLP/AITS

JEE 2025 STARS SHINE BRIGHT

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ALL INDIA OPEN CATEGORY RANKS



RUTVIK SAI
H.T.No. 256055278 (OBC-NCL)



MAJID MUJAHID HUSAIN
H.T.No. 251134112*



UJJWAL KESARI
H.T.No. 252016104*



AKSHAT KUMAR CHAURASIA
H.T.No. 254065055*

BELOW 100 ALL INDIA OPEN CATEGORY RANKS	BELOW 500 ALL INDIA OPEN CATEGORY RANKS	BELOW 1000 ALL INDIA OPEN CATEGORY RANKS	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT	NUMBER OF QUALIFIED RANKS
29	113	205	745	4,212

*DLP/AITS

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